

HR Management Platform

Work Breakdown Structure & Effort Estimates

Client Segment	Mid-size Companies (50-500 employees)
Platforms	Web Application + Mobile Application (iOS & Android)
Team	3 Developers + 1 Designer
Document Version	1.0
Date	March 10, 2026

Key Assumptions

- Team composition: 3 full-stack developers (2 senior, 1 mid-level) + 1 UI/UX designer.
- Working hours: 8 hours/day, 5 days/week. Estimates in person-days (PD).
- Tech stack: React (web), React Native (mobile), Node.js/NestJS (backend), PostgreSQL.
- Vietnamese payroll providers: integration via API with at least 2 providers (e.g., Misa, Bravo).
- Authentication via SSO/OAuth 2.0 with role-based access control (RBAC).
- Hosting on AWS (Vietnam region or Singapore for low latency).
- Mobile app shares ~60% of business logic with web via shared API layer.
- Agile methodology with 2-week sprints. Estimates include buffer for code review and QA.
- No legacy system migration in initial scope (can be added as Phase 2).
- Localization: Vietnamese (primary) + English.

Executive Summary

This project encompasses **531 person-days** of effort across **7 phases** and **21 work packages**. With a team of 3 developers working in parallel (plus 1 designer), the estimated calendar duration is approximately **8.2 months** (35.4 dev-weeks). This includes a 2-week design lead time and accounts for sequential dependencies between phases. A phased delivery approach is recommended, with the web MVP (Phases 1-3) targeting month 5, mobile app by month 7, and full launch by month 8-9.

Phase	Work Packages	Tasks	Effort (PD)	% of Total
1. Discovery & Planning	3	17	80	15.1%
2. Foundation (Backend & Auth)	3	17	56	10.5%
3. Core Modules (Web)	5	33	161	30.3%
4. Advanced Features	3	14	54	10.2%

Phase	Work Packages	Tasks	Effort (PD)	% of Total
5. Mobile Application	3	17	60	11.3%
6. Testing & Quality Assurance	2	13	80	15.1%
7. Deployment & Launch	2	11	40	7.5%
TOTAL	21	122	531	100%

Recommended Timeline

Period	Phase	Key Deliverables
Month 1-2	Discovery & Planning	Requirements, UX/UI design, architecture
Month 2-3	Foundation	Backend core, auth, web app shell
Month 3-5	Core Modules (Web)	Employee directory, org chart, leave, payroll, performance
Month 5-6	Advanced Features	Admin, dashboards, reporting, notifications
Month 5-7	Mobile App	Parallel track: React Native app development
Month 7-8	Testing & QA	Full testing cycle, UAT, bug fixes
Month 8-9	Deployment & Launch	Infrastructure, app store, go-live

Detailed Work Breakdown Structure

Phase 1: Discovery & Planning (80 person-days)

1.1 Requirements Analysis [17 PD]

WBS ID	Task	Effort (PD)	Owner
1.1.1	Stakeholder interviews & workshops	5	All
1.1.2	User persona development	3	Designer
1.1.3	Feature prioritization (MoSCoW)	2	All
1.1.4	Technical feasibility assessment	3	Dev Lead
1.1.5	Vietnamese payroll provider API research	4	Dev
Subtotal: Requirements Analysis		17	

1.2 UX/UI Design [41 PD]

WBS ID	Task	Effort (PD)	Owner
1.2.1	Information architecture & user flows	5	Designer
1.2.2	Wireframes (all core modules)	8	Designer
1.2.3	Design system / component library	6	Designer
1.2.4	Hi-fi mockups (web)	10	Designer
1.2.5	Hi-fi mockups (mobile)	7	Designer
1.2.6	Prototype & usability testing	5	Designer
Subtotal: UX/UI Design		41	

1.3 Architecture & Tech Setup [22 PD]

WBS ID	Task	Effort (PD)	Owner
1.3.1	System architecture design	4	Dev Lead
1.3.2	Database schema design	5	Dev Lead
1.3.3	API contract definition (OpenAPI)	4	Dev
1.3.4	CI/CD pipeline setup	3	Dev
1.3.5	Dev environment & tooling	3	Dev
1.3.6	Security architecture (RBAC, encryption)	3	Dev Lead
Subtotal: Architecture & Tech Setup		22	

Phase 1 Total: Discovery & Planning

80 person-days

Phase 2: Foundation (Backend & Auth) (56 person-days)

2.1 Backend Core [13 PD]

WBS ID	Task	Effort (PD)	Owner
2.1.1	Project scaffolding (NestJS)	2	Dev
2.1.2	Database setup & ORM config	3	Dev
2.1.3	Base API structure & middleware	3	Dev
2.1.4	Error handling & logging framework	2	Dev
2.1.5	File upload/storage service (S3)	3	Dev
Subtotal: Backend Core		13	

2.2 Authentication & Authorization [22 PD]

WBS ID	Task	Effort (PD)	Owner
2.2.1	User registration & login (JWT)	4	Dev
2.2.2	Role-based access control (RBAC)	5	Dev
2.2.3	SSO / OAuth integration	4	Dev
2.2.4	Password reset & account recovery	3	Dev
2.2.5	Session management & security	3	Dev
2.2.6	Audit logging	3	Dev
Subtotal: Authentication & Authorization		22	

2.3 Web App Foundation [21 PD]

WBS ID	Task	Effort (PD)	Owner
2.3.1	React project setup & config	2	Dev
2.3.2	Design system implementation	6	Dev + Designer
2.3.3	Layout & navigation shell	4	Dev
2.3.4	Auth screens (login, register, forgot pw)	4	Dev
2.3.5	State management setup	2	Dev
2.3.6	API client & error handling	3	Dev
Subtotal: Web App Foundation		21	

Phase 2 Total: Foundation (Backend & Auth)

56 person-days

Phase 3: Core Modules (Web) (161 person-days)

3.1 Employee Directory [25 PD]

WBS ID	Task	Effort (PD)	Owner
3.1.1	Employee CRUD API	5	Dev
3.1.2	Employee profile pages	5	Dev
3.1.3	Search, filter & pagination	4	Dev
3.1.4	Bulk import/export (CSV/Excel)	5	Dev
3.1.5	Employee document management	4	Dev
3.1.6	Profile photo upload & cropping	2	Dev
Subtotal: Employee Directory		25	

3.2 Organization Chart [23 PD]

WBS ID	Task	Effort (PD)	Owner
3.2.1	Org structure data model	3	Dev
3.2.2	Interactive org chart visualization	8	Dev
3.2.3	Department & team management	4	Dev
3.2.4	Drag-and-drop reorganization	5	Dev
3.2.5	Reporting line management	3	Dev
Subtotal: Organization Chart		23	

3.3 Leave Management [33 PD]

WBS ID	Task	Effort (PD)	Owner
3.3.1	Leave policy configuration engine	6	Dev
3.3.2	Leave request & approval workflow	6	Dev
3.3.3	Leave balance tracking & accrual	5	Dev
3.3.4	Team calendar view	5	Dev
3.3.5	Leave reports & analytics	4	Dev
3.3.6	Public holiday calendar (Vietnam)	3	Dev
3.3.7	Email/push notifications	4	Dev
Subtotal: Leave Management		33	

3.4 Payroll Integration [46 PD]

WBS ID	Task	Effort (PD)	Owner
3.4.1	Payroll provider API adapter layer	6	Dev
3.4.2	MISA payroll integration	8	Dev
3.4.3	Bravo payroll integration	7	Dev
3.4.4	Salary structure management	5	Dev
3.4.5	Tax calculation (Vietnamese PIT)	6	Dev
3.4.6	Social insurance (BHXH/BHYT/BHTN)	5	Dev
3.4.7	Payslip generation & distribution	5	Dev
3.4.8	Payroll reports & compliance	4	Dev
Subtotal: Payroll Integration		46	

3.5 Performance Reviews [34 PD]

WBS ID	Task	Effort (PD)	Owner
3.5.1	Review cycle configuration	5	Dev
3.5.2	Goal setting & OKR tracking	6	Dev
3.5.3	Self-assessment forms	4	Dev
3.5.4	360-degree feedback collection	6	Dev
3.5.5	Manager review & calibration	5	Dev
3.5.6	Performance analytics & dashboards	5	Dev
3.5.7	Review history & trends	3	Dev
Subtotal: Performance Reviews		34	

Phase 3 Total: Core Modules (Web)	161 person-days
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Phase 4: Advanced Features (54 person-days)

4.1 Admin & Settings [17 PD]

WBS ID	Task	Effort (PD)	Owner
4.1.1	Company settings & branding	3	Dev
4.1.2	User role management UI	4	Dev
4.1.3	Notification preferences	3	Dev
4.1.4	System configuration panel	4	Dev
4.1.5	Data backup & export tools	3	Dev
Subtotal: Admin & Settings		17	

4.2 Dashboard & Reporting [23 PD]

WBS ID	Task	Effort (PD)	Owner
4.2.1	Executive dashboard	5	Dev
4.2.2	HR analytics & KPIs	5	Dev
4.2.3	Custom report builder	6	Dev
4.2.4	Export to PDF/Excel	4	Dev
4.2.5	Scheduled report delivery	3	Dev
Subtotal: Dashboard & Reporting		23	

4.3 Notifications & Communication [14 PD]

WBS ID	Task	Effort (PD)	Owner
4.3.1	Email notification engine	4	Dev
4.3.2	In-app notification center	4	Dev
4.3.3	Push notification service	3	Dev
4.3.4	Announcement board	3	Dev
Subtotal: Notifications & Communication		14	

Phase 4 Total: Advanced Features

54 person-days

Phase 5: Mobile Application (60 person-days)

5.1 Mobile Foundation [19 PD]

WBS ID	Task	Effort (PD)	Owner
5.1.1	React Native project setup	3	Dev
5.1.2	Mobile design system implementation	5	Dev + Designer
5.1.3	Navigation & auth flow	4	Dev
5.1.4	Offline data caching strategy	4	Dev
5.1.5	Push notification integration	3	Dev
Subtotal: Mobile Foundation		19	

5.2 Mobile Core Features [27 PD]

WBS ID	Task	Effort (PD)	Owner
5.2.1	Employee directory (mobile)	5	Dev
5.2.2	Leave request & approval (mobile)	5	Dev
5.2.3	Payslip viewer (mobile)	3	Dev
5.2.4	Org chart viewer (mobile)	4	Dev
5.2.5	Performance review (mobile)	4	Dev
5.2.6	Notification center (mobile)	3	Dev
5.2.7	Profile management (mobile)	3	Dev
Subtotal: Mobile Core Features		27	

5.3 Mobile Platform-Specific [14 PD]

WBS ID	Task	Effort (PD)	Owner
5.3.1	iOS-specific optimizations	3	Dev
5.3.2	Android-specific optimizations	3	Dev
5.3.3	Biometric authentication	3	Dev
5.3.4	Deep linking	2	Dev
5.3.5	App store preparation	3	Dev
Subtotal: Mobile Platform-Specific		14	

Phase 5 Total: Mobile Application

60 person-days

Phase 6: Testing & Quality Assurance (80 person-days)

6.1 Testing [48 PD]

WBS ID	Task	Effort (PD)	Owner
6.1.1	Unit test coverage (backend)	8	Dev
6.1.2	Unit test coverage (frontend)	6	Dev
6.1.3	Integration testing (API)	6	Dev
6.1.4	End-to-end testing (Cypress/Playwright)	8	Dev
6.1.5	Mobile testing (both platforms)	6	Dev
6.1.6	Performance & load testing	4	Dev
6.1.7	Security testing & penetration test	5	Dev
6.1.8	UAT preparation & support	5	All
Subtotal: Testing		48	

6.2 Bug Fixing & Polish [32 PD]

WBS ID	Task	Effort (PD)	Owner
6.2.1	Bug triage & fixing (web)	10	Dev
6.2.2	Bug triage & fixing (mobile)	8	Dev
6.2.3	UI polish & design QA	5	Dev + Designer
6.2.4	Performance optimization	5	Dev
6.2.5	Accessibility audit & fixes	4	Dev + Designer
Subtotal: Bug Fixing & Polish		32	

Phase 6 Total: Testing & Quality Assurance	80 person-days
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Phase 7: Deployment & Launch (40 person-days)

7.1 Infrastructure & DevOps [16 PD]

WBS ID	Task	Effort (PD)	Owner
7.1.1	Production infrastructure setup (AWS)	5	Dev
7.1.2	Database migration & seeding	3	Dev
7.1.3	SSL, DNS & domain config	2	Dev
7.1.4	Monitoring & alerting (Datadog/CloudWatch)	3	Dev
7.1.5	Backup & disaster recovery	3	Dev
Subtotal: Infrastructure & DevOps		16	

7.2 Launch Activities [24 PD]

WBS ID	Task	Effort (PD)	Owner
7.2.1	App store submission (iOS + Android)	3	Dev
7.2.2	User documentation & help center	5	All
7.2.3	Admin training materials	3	All
7.2.4	Data migration tooling	5	Dev
7.2.5	Go-live checklist & execution	3	All
7.2.6	Post-launch monitoring & hotfixes	5	Dev
Subtotal: Launch Activities		24	

Phase 7 Total: Deployment & Launch	40 person-days
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Effort Distribution by Role

Role	Person-Days	Percentage	Notes
Developers (3)	444	83.6%	Split across 3 devs for parallelism
Designer (1)	44	8.3%	Front-loaded in Phase 1, advisory later
Dev + Designer (shared)	20	3.8%	Collaborative tasks (design system, QA)
All Team	23	4.3%	Workshops, UAT, launch activities
Total	531	100%	

Risk Register

Risk	Impact	Mitigation	Phase
Vietnamese payroll API instability	High	Payroll providers may have limited/undocumented APIs. Build adapter layer with fallback modes.	3.4
Scope creep on performance module	Medium	Performance reviews can expand indefinitely. Lock scope to defined review types.	3.5
React Native performance on low-end devices	Medium	Vietnamese market includes budget Android phones. Profile early and set device baseline.	5.x
Regulatory changes (tax/insurance)	Medium	Vietnamese labor law changes frequently. Build configurable rule engine, not hardcoded.	3.4
Team capacity (3 devs is lean)	High	Parallel workstreams limited. Sequence phases carefully. Consider contractor for payroll.	All
Data privacy compliance	Medium	Vietnam Cybersecurity Law requirements. Engage legal counsel early for data residency.	2.2

Cost Estimation Guidance

The following table provides indicative cost ranges based on Vietnamese market rates for the team composition specified. Actual costs will vary based on seniority, location (HCMC/Hanoi vs. other cities), and engagement model.

Role	Qty	Rate Range (USD/month)	Duration	Est. Cost Range
Senior Developer	2	\$2,500 - \$4,000	9 months	\$45,000 - \$72,000
Mid-level Developer	1	\$1,500 - \$2,500	9 months	\$13,500 - \$22,500
UI/UX Designer	1	\$1,500 - \$3,000	6 months*	\$9,000 - \$18,000
Infrastructure (AWS)	-	\$500 - \$1,500/mo	12 months	\$6,000 - \$18,000
Estimated Total				\$73,500 - \$130,500

* Designer full-time for Phase 1, part-time advisory for remaining phases.

Recommendations

- **Phased delivery:** Ship Web MVP (directory + leave + org chart) by month 5 for early feedback, then iterate.
- **Payroll integration:** Start with one provider (MISA) first; add Bravo in a subsequent sprint. This is the highest-risk module.
- **Mobile priority:** Begin mobile development in parallel with Phase 3 (month 3) to avoid becoming a bottleneck.
- **Contractor consideration:** With 3 developers, parallel capacity is limited. Consider a contract developer for payroll integration specifically.
- **Design lead time:** Designer should be 2-4 weeks ahead of development. Wireframes and mockups should be approved before sprint planning.
- **Testing strategy:** Invest in automated E2E tests early (Phase 2). Manual QA for payroll calculations is critical given Vietnamese tax complexity.
- **Security audit:** Schedule an external security review before launch, especially for payroll/PII data handling.
- **Localization:** Build i18n infrastructure from day one. Vietnamese primary, English secondary. Content translation is a parallel workstream.